

Range of a Function

Jonathan R. Bacolod

Function

A **function** assigns exactly one output to each input.

Range

The **range** of a function is the set of all output values (y -values).

Range from Ordered Pairs

In a set of ordered pairs, the range is obtained by listing all second coordinates.

Range from a Table

In a table of values, the range consists of all values in the output column.

Range from a Mapping Diagram

In a mapping diagram, the range is the set of values that receive arrows from the domain.

Range from a Graph

In a graph, the range is the set of all y -values covered by the graph.

Range from an Equation

In an equation, substitute the given domain values, if provided, and determine the corresponding outputs.

Examples

Example 1

Determine the **range** of

$$\{(1, 4), (2, 6), (3, 8), (4, 10)\}.$$

$$\text{Range} = \{4, 6, 8, 10\}$$

Example 2

Determine the **range** of

$$\{(-2, 5), (-1, 3), (0, 1), (1, -1)\}.$$

$$\text{Range} = \{5, 3, 1, -1\}$$

Example 3

Determine the **range** of the mapping:

$$1 \rightarrow 2, \quad 2 \rightarrow 4, \quad 3 \rightarrow 6.$$

$$\text{Range} = \{2, 4, 6\}$$

Example 4

Determine the **range** of

$$y = 2x + 1$$

for the domain $\{0, 1, 2, 3, 4\}$.

$$\text{Range} = \{1, 3, 5, 7, 9\}$$

Practice Exercises

Practice 1

Determine the **range** of

$$\{(1, 4), (2, 6), (3, 8), (4, 10)\}.$$

Practice 2

Determine the **range** of

$$\{(-2, 5), (-1, 3), (0, 1), (1, -1)\}.$$

Practice 3

Determine the **range** of the mapping:

$$1 \rightarrow 2, \quad 2 \rightarrow 4, \quad 3 \rightarrow 6.$$

Practice 4

Determine the **range** of the mapping:

$$-1 \rightarrow 1, \quad 0 \rightarrow 0, \quad 1 \rightarrow 1, \quad 2 \rightarrow 4.$$

Practice 5

Determine the **range** of

$$y = 2x + 1$$

for the domain $\{0, 1, 2, 3, 4\}$.

Practice 6

Determine the **range** of

$$y = x^2 - 1$$

for the domain $\{-2, -1, 0, 1, 2\}$.

Practice 7

Determine the **range** of

$$\{(-2, -3), (-1, -1), (0, 1), (1, 3), (2, 5)\}.$$

Practice 8

Determine the **range** of

$$\{(-3, 4), (-2, 2), (-1, 0), (0, -2), (1, -4)\}.$$

Activities

Activity 1

Determine the **range** of

$$\{(2, 5), (3, 7), (4, 9), (5, 11)\}.$$

Activity 2

Determine the **range** of

$$\{(-3, 8), (-2, 6), (-1, 4), (0, 2)\}.$$

Activity 3

Determine the **range** of the mapping:

$$2 \rightarrow 3, \quad 3 \rightarrow 6, \quad 4 \rightarrow 9.$$

Activity 4

Determine the **range** of the mapping:

$$-2 \rightarrow 4, \quad -1 \rightarrow 1, \quad 0 \rightarrow 0, \quad 1 \rightarrow 1.$$

Activity 5

Determine the **range** of

$$y = 3x - 2$$

for the domain $\{1, 2, 3, 4, 5\}$.

Activity 6

Determine the **range** of

$$y = x^2 + 2$$

for the domain $\{-2, -1, 0, 1, 2\}$.

Activity 7

Determine the **range** of

$$\{(-2, -4), (-1, -2), (0, 0), (1, 2), (2, 4)\}.$$

Activity 8

Determine the **range** of

$$\{(-2, 6), (-1, 3), (0, 0), (1, -3), (2, -6)\}.$$

Thank you!